THE UNIVERSITY OF LAHORE

Faculty of Information Technology | Department of Software Engineering

Mobile Application Development (CS02310) — Lab Task No. 02 Report

UOL Course Catalogue — RecyclerView List App

Student Name:	**Faseeh Ur Rehman**	**Course Code:**	CS02310 (Mobile App Dev)
SAP ID:	**70147860**	**Programme & Sem:**	BS Software Engineering (7th Sem)
Section:	**Section S**	**Mapped CLO:**	CLO-4 (Bloom's: C6 / Modern Tool Usage)
GitHub Repo:	https://github.com/Leetfraud/LabTask2_RecyclerView_70147860	**Term:**	Fall 2026
GitHub Profile:	https://github.com/Leetfraud		

Scenario 1, Step 5 — Running the List Without setLayoutManager()

What I saw: with the setLayoutManager() line removed, the app opened but the list area was completely empty — no course cards were shown at all.
Why: a RecyclerView does not position items by itself. The LayoutManager decides how and where each card is measured and placed, so without one the RecyclerView never asks the adapter for any cards and draws nothing. After restoring rvCourses.setLayoutManager(new LinearLayoutManager(this)), the vertical list came back.

Scenario 2, Step 5 — Why the Position Is Used to Fetch the Data

The position passed to onItemClick() is the index of the tapped card in the adapter, and it is the same index as that course in the courses ArrayList, so courses.get(position) returns exactly the tapped course.
Card views are recycled and reused for different courses while scrolling, so the view itself cannot identify the course — only the position reliably does, which is why only that course's data is sent.

Step-by-Step Implementation Summary

Scenario 1 — The Course List

- **Step 1 (Model class):** Course has private fields code, title, instructor, email, description (String) and creditHours (int), one constructor that sets every field, and a public getter for each.
- **Step 2 (Card layout):** item_course.xml has a CardView as its root with app:cardCornerRadius="12dp" and app:cardElevation="4dp", and three TextViews for the course code, title and credit hours.
- **Step 3 (Adapter):** CourseAdapter extends RecyclerView.Adapter. Its static inner ViewHolder finds the three TextViews once in its constructor. onCreateViewHolder() inflates the card, onBindViewHolder() writes one course's data into it, and getItemCount() returns the list size.
- **Step 4 (Data and LayoutManager):** MainActivity builds an ArrayList of ten BS-SE courses, sets a LinearLayoutManager on rvCourses and attaches the adapter with setAdapter().
- **Step 5 (List test):** scrolled through all ten courses, then ran the app once without setLayoutManager() (observation above, Figure 2) and restored the call.

Scenario 2 — The Course Details Screen

- **Step 1 (Click interface):** CourseAdapter declares the interface OnItemClickListener with onItemClick(int position). The ViewHolder constructor attaches a click listener to itemView that reports getBindingAdapterPosition(), and MainActivity implements the interface.
- **Step 2 (Explicit Intent):** onItemClick() reads the tapped course with courses.get(position) and opens CourseDetailActivity with an explicit Intent, sending only that course's data through putExtra() with public static final key constants (EXTRA_CODE, EXTRA_TITLE, etc.).
- **Step 3 (Details screen):** CourseDetailActivity is declared in AndroidManifest.xml. It reads the values with getStringExtra() and getIntExtra(), shows the title, instructor, credit hours and description, and calls setTitle(code) so the screen title is the course code.
- **Step 4 (Email Instructor):** the button builds an ACTION_SENDTO Intent with "mailto:" + email, so the email app opens addressed to that course's instructor only. startActivity() is inside try–catch and a Toast is shown on ActivityNotFoundException.
- **Step 5 (Item click test):** tapped three different courses; each opened a details screen showing only the tapped course: SE-3102, SE-3205 and SE-4101 (Figures 3–5). Tapping "Email Instructor" on SE-3301 opened Gmail addressed only to nida.farooq@se.uol.edu.pk (Figure 6).

App Screenshots

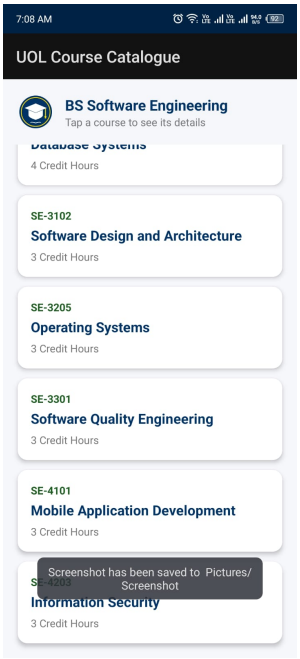


Figure 1: Course List

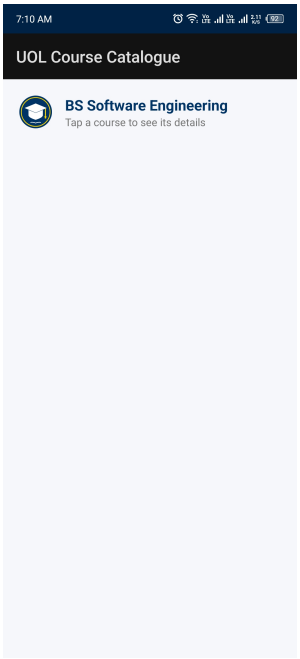


Figure 2: Without setLayoutManager()

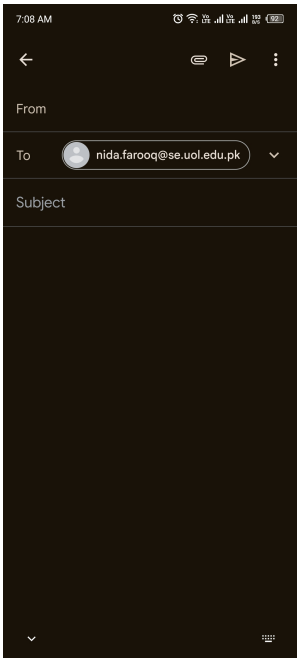


Figure 6: Email app (SE-3301 instructor)

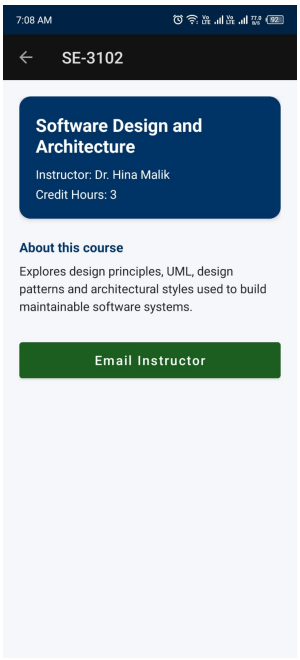


Figure 3: Details — SE-3102

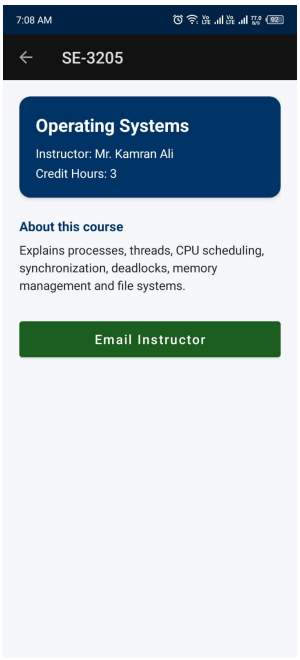


Figure 4: Details — SE-3205

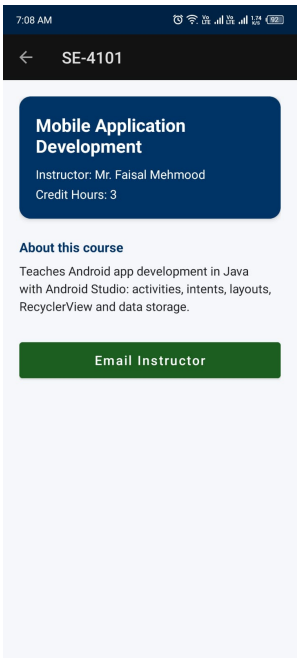


Figure 5: Details — SE-4101